How-To Library

Informatica

Address Verification Best Practices for Spain

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Abstract

This document tells you how to connect the elements in a Spain address to the appropriate address elements in Informatica Address Verification. This document also describes best practices that you can implement when you verify addresses in Spain.

Supported Versions

- Informatica Address Verification 3.0 and later
- Informatica Address Verification (On-Premises) 5.x

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Introduction

Informatica Address Verification can validate and enhance addresses from Spain. This how-to article provides you with configuration information and best practices that you can follow to get the most from the addresses. The best practices include advice on how to select the appropriate XML elements for Spain address data.

To verify the quality of an input address, Address Verification compares the address information to the data in one or more reference data files. You can configure Address Verification to identify the different types of address information in the input data.

When you select the input address elements, you must ensure that the elements are correct for the different types of input information. The country that the address identifies determines the elements that you need.

An address in Spain can contain different types of information, including street information and locality information. You must map each item of address information to an appropriate element in Address Verification.

Process Modes

Informatica Address Verification supports the following process modes for Spain addresses:

- BATCH
- COUNTRY RECOGNITION
- FAST COMPLETION

- INTERACTIVE
- PARSE

Character Set

Spain reference address databases store address data in the Latin-1 script. By default, Informatica Address Verification returns the verified Spain addresses in the Latin-1 script.

Address Format and Address Structure

You improve the performance of Address Verification when you define the input address in the format that the postal service expects.

The Spain postal service recognizes the following structure for a postal address:

```
Line 1: RECIPIENT OR CONTACT NAME
Line 2: [URBANIZATION]
Line 3: STREET TYPE STREET NAME HOUSE OR BUILDING NUMBER [SUB-BUILDING INFORMATION]
Line 4: POSTAL CODE CITY
Line 5: COUNTRY
```

The following sample address illustrates the structure:

```
RUA ALCALDE ABELLA 23 ESCALERA 1 PLANTA -1 PUERTA A
15002 A CORUÑA
SPAIN
```

Key Address Elements

Address Verification is unlikely to verify an address successfully if one of the following items is absent from the address:

- House Number
- Street
- Locality
- Postal Code

Mapping Input Address Data to Address Elements

To verify an address, map the input address information to appropriate elements in Address Verification.

The following table shows how you can map the input address to the address elements:

Information Type	Address Information	Address Elements
Street Type, Street Name	RUA ALCALDE ABELLA	Street
House Number	23	HouseNumber
Urbanization	Not used in the current address	Locality_2
Staircase, Floor, Door	ESCALERA 1 PLANTA -1 PUERTA A	Sub-building
Postal Code	15002	PostalCode

Information Type	Address Information	Address Elements
City	A CORUÑA	Locality_1
Country	SPAIN	Country

Input and Result XML in Address Verification (On-Premises)

The following example shows how you can map the address information to the XML address elements in Address Verification (On-Premises).

You enter the following input address:

```
<InputData>
    <AddressElements>
        <Country Item="1" Type="NAME">SPAIN</Country>
        <Locality Item="1" Type="COMPLETE">A CORUÑA</Locality>
        <PostalCode Item="1" Type="UNFORMATTED">15002</PostalCode>
        <Street Item="1" Type="COMPLETE">RUA ALCALDE ABELLA</Street>
        <Number Item="1" Type="COMPLETE">23</Number>
        <SubBuilding Item="1" Type="COMPLETE">ESCALERA 1 PLANTA -1 PUERTA A</SubBuilding>
    </AddressElements>
</InputData>
```

Informatica Address Verification (On-Premises) returns the following result:

```
<?xml version="1.0" encoding="UTF-16"?>

<!-- EngineVersion="5.11.0.41659" -->

<Result     ProcessStatus="V4"
     ModeUsed="INTERACTIVE"
     Count="1"
     CountOverflow="NO"
     CountryISO3="ESP"
     PreferredScript="LATIN_1"
     PreferredLanguage="ENGLISH">

<ResultData ResultNumber="1"
     MailabilityScore="5"
     ResultPercentage="100.00"
     ElementResultStatus="F0F080F0F00000EF00E0"
     ElementInputStatus="60600060600000560060"
     AddressResolutionCode="00000000000000000000"
     ExtElementStatus="00000000000000060000"
     ElementRelevance="10100010100000110010"
     AddressType="S"
     LanguageISO3="SPA">
    <AddressElements>
        <Country Type="NAME_EN" Item="1">Spain</Country>
        <Locality Item="1">A Coruña</Locality>
        <PostalCode Item="1">15002</PostalCode>
        <Province Item="1">A Coruña</Province>
        <Street Item="1">Rua Alcalde Abella</Street>
        <Number Item="1">23</Number>
        <SubBuilding Item="1">Escalera 1</SubBuilding>
        <SubBuilding Item="2">P. -1</SubBuilding>
        <SubBuilding Item="3">Pta. A</SubBuilding>
    </AddressElements>
    <AddressLines>
        <DeliveryAddressLine Line="1">Rua Alcalde Abella 23 Escalera 1 P. -1 Pta. A</
DeliveryAddressLine>
        <CountrySpecificLocalityLine Line="1">15002 A Coruña</CountrySpecificLocalityLine>
        <FormattedAddressLine Line="1">Rua Alcalde Abella 23 Escalera 1 P. -1 Pta. A</
FormattedAddressLine>
        <FormattedAddressLine Line="2">15002 A Coruña</FormattedAddressLine>
        <FormattedAddressLine Line="3">Spain</FormattedAddressLine>
```

```
    </AddressLines>
    <AddressComplete>Rua Alcalde Abella 23 Escalera 1 P. -1 Pta. A
15002 A Coruña
Spain</AddressComplete>
</ResultData>
</Result>
```

Input and Result XML in Address Verification

The following example shows how you can verify a Spain address in a SOAP call to Address Verification.

You submit the following request:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope" xmlns:v="http://
validator5.AddressDoctor.com/Webservice5/v4">
   <soap:Header/>
   <soap:Body>
      <v:Process>
         <v:login>*****</v:login>
         <v:password>********</v:password>
         <v:parameters>
            <v:ProcessMode>BATCH</v:ProcessMode>
            <v:ServiceParameters></v:ServiceParameters>
            <v:ValidationParameters></v:ValidationParameters>
         </v:parameters>
         <v:addresses>
            <v:Address>
               <v:SubBuilding>
                  <v:string>ESCALERA 1</v:string>
                  <v:string>PLANTA -1</v:string>
                  <v:string>PUERTA A</v:string>
               </v:SubBuilding>
               <v:Street>
                  <v:string>RUA ALCALDE ABELLA</v:string>
               </v:Street>
              <v:HouseNumber>
                  <v:string>23</v:string>
               </v:HouseNumber>
               <v:Locality>
                  <v:string>A CORUÑA</v:string>
               </v:Locality>
               <v:PostalCode>
                  <v:string>15002</v:string>
               </v:PostalCode>
               <v:Province>
                  <v:string>A CORUÑA</v:string>
               </v:Province>
               <v:Country>
                  <v:string>SPAIN</v:string>
               </v:Country>
            </v:Address>
         </v:addresses>
         <v:enrichments></v:enrichments>
      </v:Process>
   </soap:Body>
</soap:Envelope>
```

Informatica Address Verification returns the following response:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
   <soap:Body>
      <ProcessResponse xmlns="http://validator5.AddressDoctor.com/Webservice5/v4">
         <ProcessResult>
            <StatusCode>100</StatusCode>
            <StatusMessage>OK</StatusMessage>
            <Results>
               <Result>
                  <ProcessStatus>V4</ProcessStatus>
                  <CountryISO3>ESP</CountryISO3>
                  <ResultDataSet>
                     <ResultData>
```

```
                        <ResultNumber>1</ResultNumber>
                        <MailabilityScore>5</MailabilityScore>
                        <ResultPercentage>100.00</ResultPercentage>
                        <ElementInputStatus>6060606060000560060</ElementInputStatus>
                        <ElementResultStatus>F0F0F0F0F00000EF00E0</ElementResultStatus>
                        <ElementRelevance>1010001010000110010</ElementRelevance>
                        <ExtElementStatus>0000000000000060000</ExtElementStatus>
                        <AddressResolutionCode>00000000000000000000</AddressResolutionCode>
                        <AddressType>S</AddressType>
                        <LanguageISO3>SPA</LanguageISO3>
                        <Address>
                           <SubBuilding>
                              <string>ESCALERA 1</string>
                              <string>PLANTA -1</string>
                              <string>PUERTA A</string>
                           </SubBuilding>
                           <Street>
                              <string>RUA ALCALDE ABELLA</string>
                           </Street>
                           <HouseNumber>
                              <string>23</string>
                           </HouseNumber>
                           <Locality>
                              <string>A CORUÑA</string>
                           </Locality>
                           <PostalCode>
                              <string>15002</string>
                           </PostalCode>
                           <Province>
                              <string>A CORUÑA</string>
                           </Province>
                           <Country>
                              <string>SPAIN</string>
                           </Country>
                           <DeliveryAddressLines>
                              <string>RUA ALCALDE ABELLA 23 ESCALERA 1 PLANTA -1 PUERTA A</
string>
                           </DeliveryAddressLines>
                           <CountrySpecificLocalityLine>
                              <string>15002 A CORUÑA</string>
                           </CountrySpecificLocalityLine>
                           <FormattedAddress>
                              <string>RUA ALCALDE ABELLA 23 ESCALERA 1 PLANTA -1 PUERTA A</
string>
                              <string>15002 A CORUÑA</string>
                              <string>SPAIN</string>
                           </FormattedAddress>
                           <AddressComplete>RUA ALCALDE ABELLA 23 ESCALERA 1 PLANTA -1 PUERTA A;
15002 A CORUÑA;SPAIN</AddressComplete>
                        </Address>
                     </ResultData>
                  </ResultDataSet>
               </Result>
            </Results>
         </ProcessResult>
      </ProcessResponse>
   </soap:Body>
</soap:Envelope>
```


Address Enrichments

You can configure Informatica Address Verification to add enrichments to verified Spain addresses.

The following table lists the type of enrichments that you can add to a Spain address:

Enrichment Type	Description
CAMEO	You can retrieve CAMEO profiles as enrichments for valid Spain addresses. CAMEO profiles are consumer classification systems that provide socio-economic and demographic indicators about the residents at the address.
Country-Specific Address Enrichments	You can retrieve the following enrichments for valid Spain addresses: - INE_PROVINCE_CODE - INE_MUNICIPALITY_CODE - INE_STREET_CODE The INE codes contains the series of values that identify the province, municipality, and street to which the address belongs.
Geocoding	You can retrieve the standard, interpolated geocoordinates for valid Spain addresses.

Address Verification Tips and Best Practices

Note the following points when you verify Spain addresses:

- You can abbreviate the Calle (street) information as `C/` or `C` in a Spain address.

  For example, the following address uses `C/` for Calle:

  ```
  C/ del Pobo 29
  28816 Valdeavero
  Spain
  ```

- The house number information in a Spain address can contain letters of the alphabet.

  For example, the house number in the following address contains the letter `B`:

  ```
  Calle Don Pedro 11B0
  02600 Villarrobledo
  Spain
  ```

- An urbanization name can assist in address verification if the address does not contain a valid postcode or city name. Address Verification maps the urbanization name to the Locality_2 address element.
- Address Verification does not support address information in FormattedAddressLine or DeliveryAddressLine elements in fast completion mode for Spain addresses.
- Spain addresses can contain multiple terms and abbreviations.

  The following table lists a range of terms or abbreviations and the corresponding descriptions:

Term or Abbreviation	Description
Avda.	Avenue
Barrio	Neighborhood of a city
C	Calle (Street)
Calle	Street
Casilla de Correo	Post Office Box

Term or Abbreviation	Description
Dpto. Or Dto.	Department
No.	Precedes a house number or building number in some addresses.
Piso Or Planta	Floor
Placita	Place
Pta.	Puerta (Door)
s/n	Without (sin) number, for example when a building on a street does not have a building number or house number.

Frequently Asked Questions

How does Address Verification handle Isles information in Spain?

Address Verification returns the isles name in the Province element. The province name is not an essential component of a Spain address.

How does Address Verification handle province or state (region, community, and autonomous city) information in Spain?

Address Verification returns the information in the Province element. The province name is not an essential component of a Spain address.

Can a locality contain more than one postal code in a Spain address?

Yes. For example, the locality `Coruña` contains postal codes in the range 15001 through 15007.

Can a postal code represent more than one locality in a Spain address?

Yes. For example, the postal code 15008 represents eight localities.

In what format does Address Verification return the province code?

Address Verification returns the province code according to the ISO 3166-2:ES standard. For example, the province code for Madrid is `ES-MD`.

The province code is not an essential component of a Spain address. To return the code, use the Province Abbreviation element.

How does Address Verification process address element descriptors in a Spain address?

Address Verification returns the address element descriptors in the regional language. For example, Address Verification returns `carrer` (Catalan) in place of `calle` (Spanish) as a street descriptor.

Does Address Verification recognize an address that includes the SN (sin número) house number indicator?

Yes. If a valid street address does not include a house number, and if you include SN in the street information, Address Verification parses SN as an indicator that the address does not use a house number.

Reference Address Databases

Informatica updates the Spain reference address databases quarterly. You can download the latest database packages from the Informatica Database Download portal. For news about reference address databases and software updates for Address Verification, subscribe to the UpData newsletter from Informatica.

Informatica Address Verification provides the following reference databases for Spain addresses:

- `ESP5BI.MD`. Batch and interactive modes.
- `ESP5FC.MD`. Fast completion mode.
- `ESP5GC.MD`. Standard geocoding.
- `ESP5E1.MD`. Supplementary data.
- `ESP5CA.MD`. CAMEO.

Note: You download databases for Informatica Address Verification (On-Premises). You do not need to download databases for Informatica Address Verification, as Informatica Address Verification runs on Informatica Cloud.

Additional Documentation Resources

To view the latest documentation for Informatica Address Verification, click the following links:

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor-cloud

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